

AnalystLab Africa – Week 1–2 Summary Report

Data Cleaning & EDA: E-commerce and Netflix Datasets

Cleaning Challenges

For the E-commerce dataset, the biggest challenge was the large number of missing CustomerID values (about 133,000 rows). Since deleting all of them would remove a quarter of the data, I decided to leave them as they are, since they likely represent guest checkouts and don't affect product or country-level analysis. I also found 5,268 duplicate rows and 1,058 rows with invalid (zero or negative) prices, which I removed. I separated returns (negative quantities) into their own table instead of deleting them, since they are still valid business data.

For the Netflix dataset, the main challenge was missing values in the director, cast, and country columns. Instead of deleting these rows, I filled them with "Unknown" so I wouldn't lose useful information. Rows with missing dates, ratings, or duration were removed since they were very few.

Key EDA Findings

For the E-commerce dataset, the United Kingdom generates the vast majority of revenue, and sales peak towards the end of the year, likely due to holiday shopping. A small number of customers are responsible for a large share of total spending.

For the Netflix dataset, Movies make up about 70% of the content, and content additions grew rapidly from 2016 to 2019. The United States is the largest content producer, and most content is rated for mature or teen audiences (TV-MA and TV-14).

Top Insights

1. The E-commerce business relies heavily on the UK market and could benefit from expanding into other countries.
2. A small group of high-value customers drive a large share of revenue, so a loyalty program could help retain them.
3. Netflix's content library is dominated by adult-oriented content, with fewer options for children and families.
4. Both datasets show clear growth trends over time, which businesses can use to plan ahead (e.g., stocking up before peak sales months, or adding more content during growth periods).